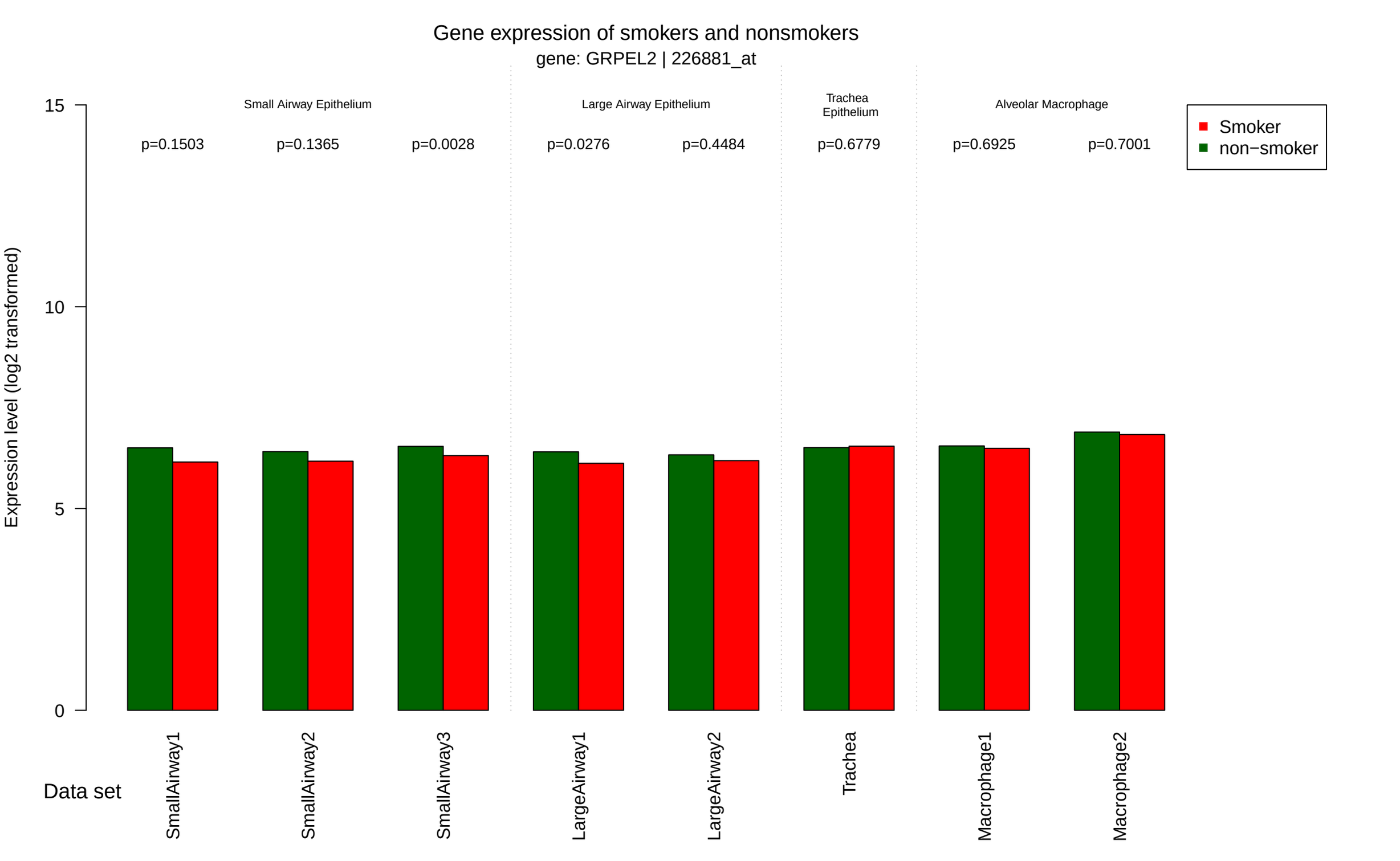
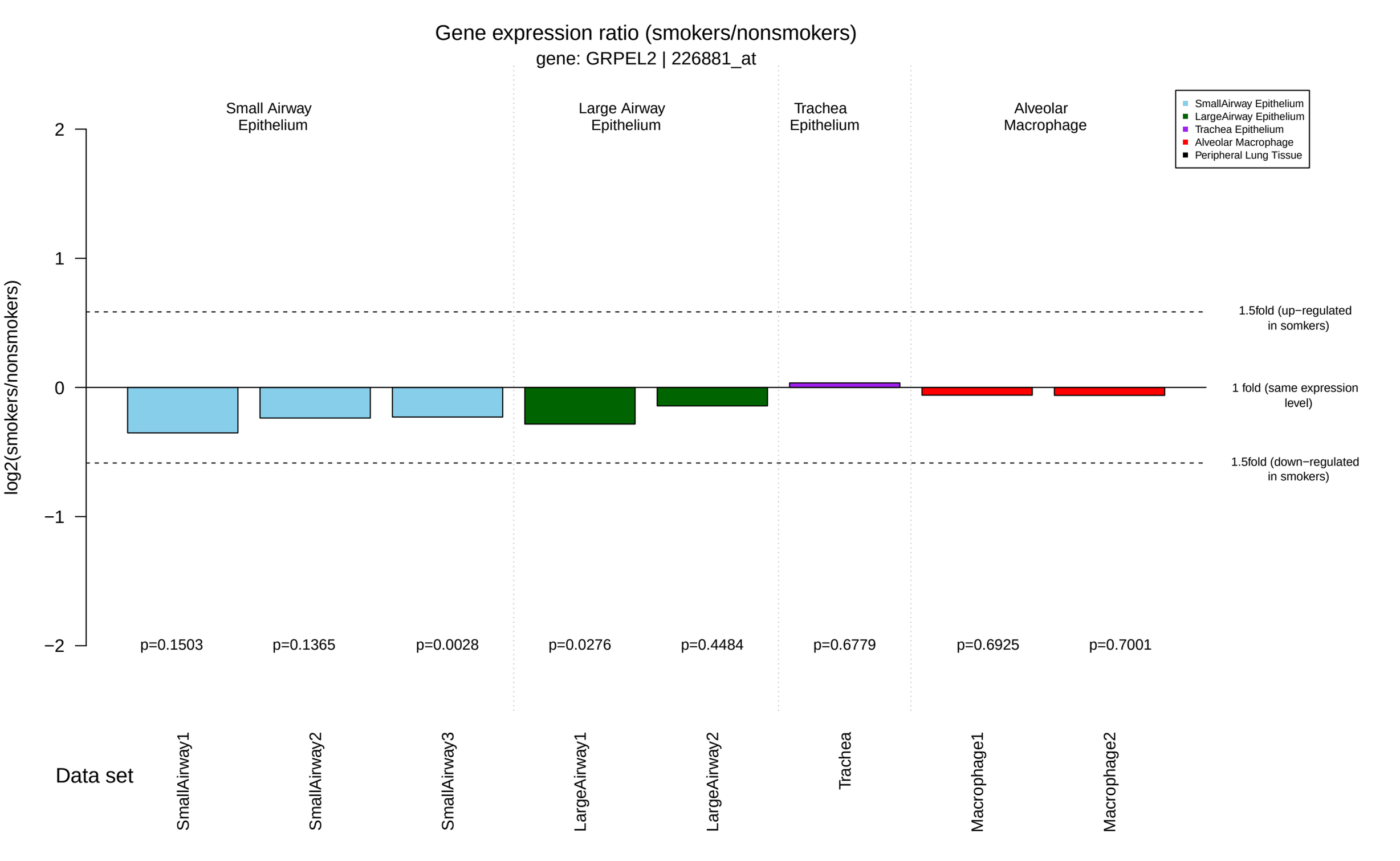
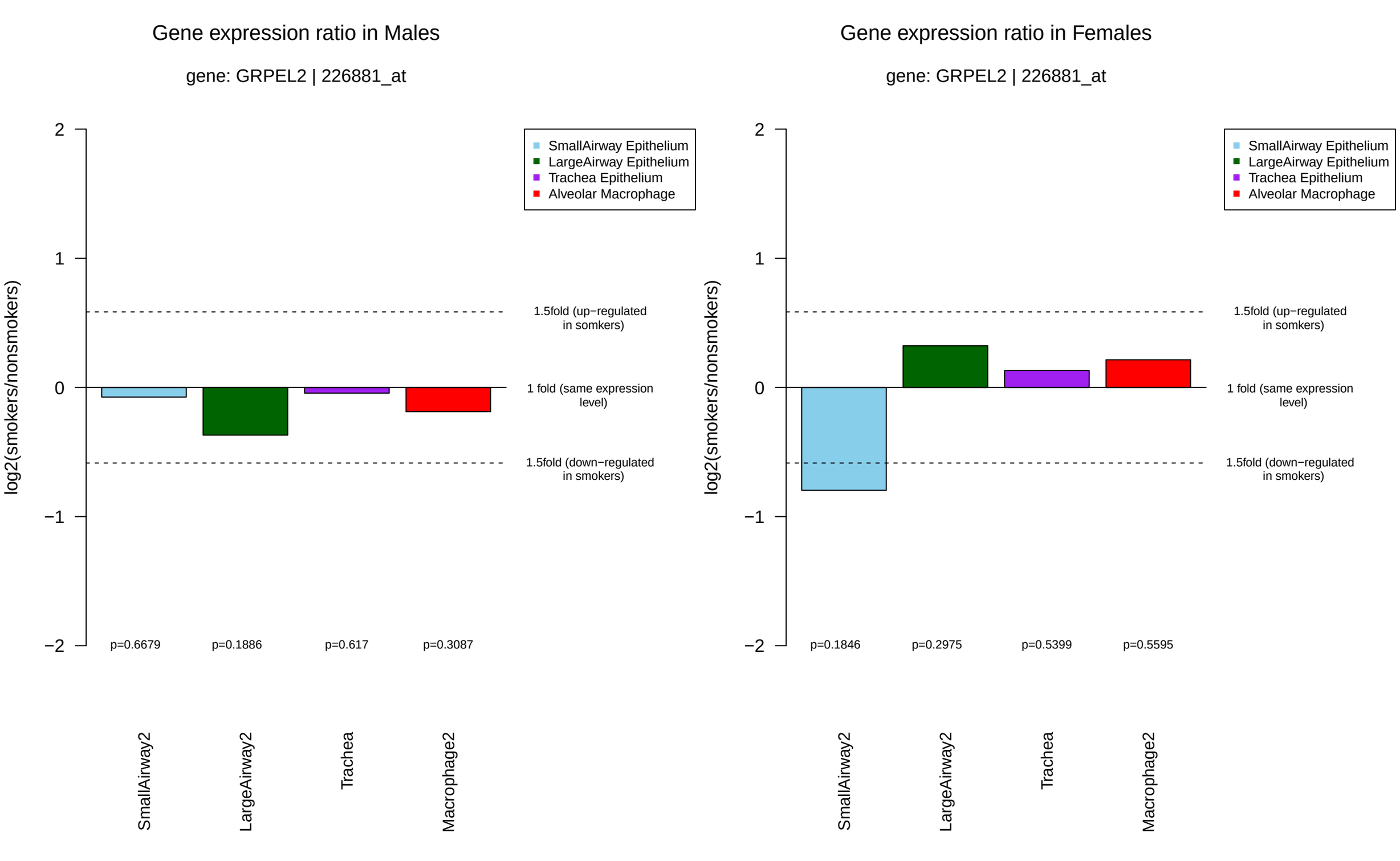
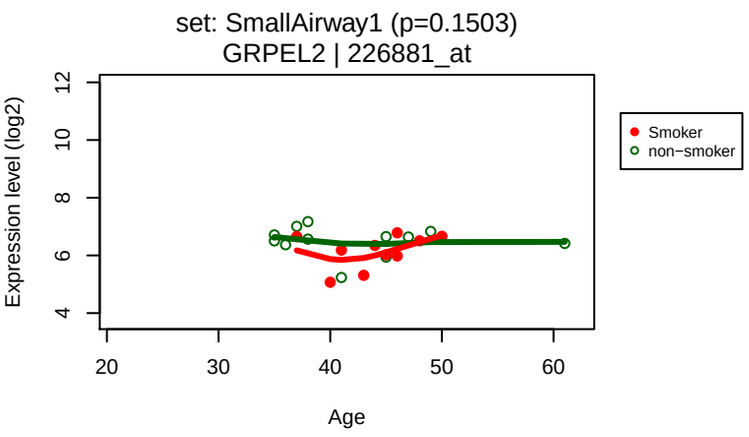
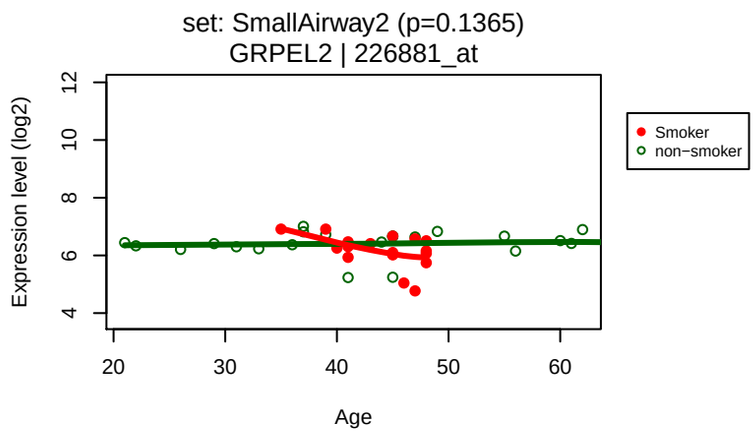
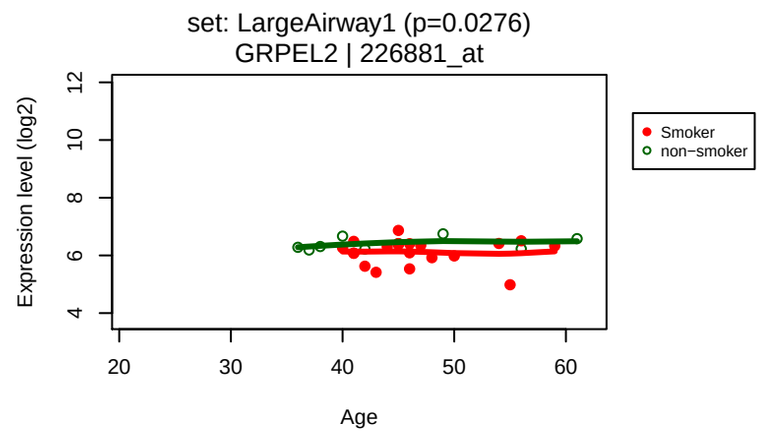

Gene expression of smokers and nonsmokers
gene: GRPEL2 | 226881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1503
p=0.1365
p=0.0028
p=0.0276
p=0.4484
p=0.6779
p=0.6925
p=0.7001
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

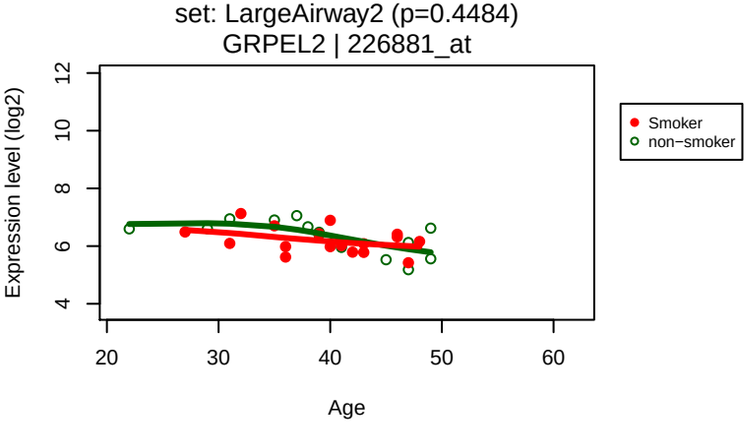

Gene expression ratio (smokers/nonsmokers)
gene: GRPEL2 | 226881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1503
p=0.1365
p=0.0028
p=0.0276
p=0.4484
p=0.6779
p=0.6925
p=0.7001
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

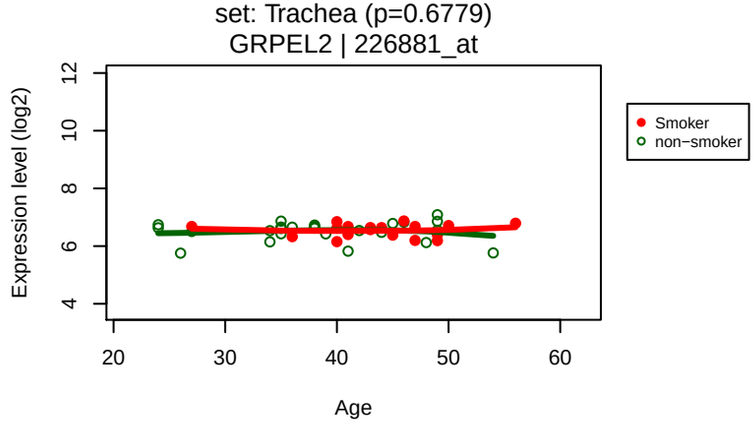

Gene expression ratio in Males
gene: GRPEL2 | 226881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6679
p=0.1886
p=0.617
p=0.3087
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRPEL2 | 226881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1846
p=0.2975
p=0.5399
p=0.5595
SmallAirway2
LargeAirway2
Trachea
Macrophage2

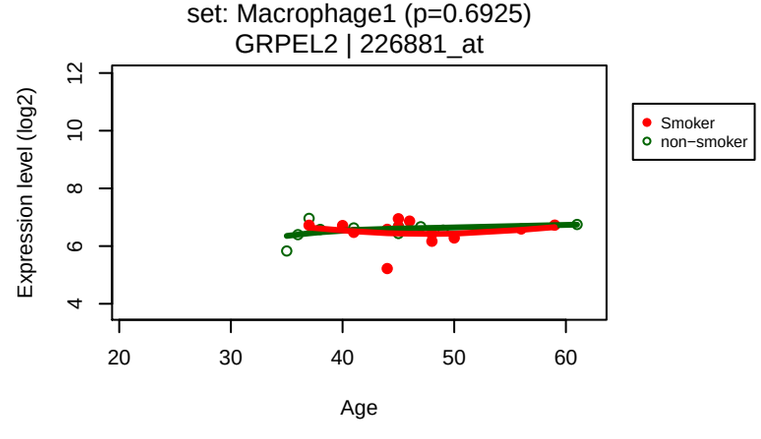

set: SmallAirway1 (p=0.1503)
GRPEL2 | 226881_at
Expression level (log2)
Age
Smoker
non−smoker

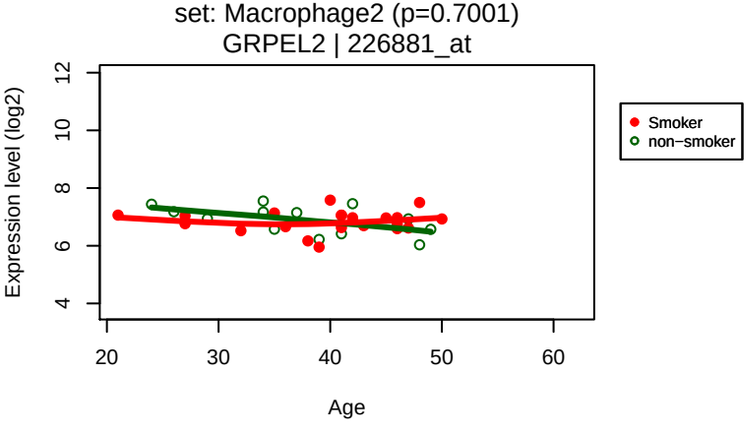

set: SmallAirway2 (p=0.1365)
GRPEL2 | 226881_at
Expression level (log2)
Age
Smoker
non−smoker

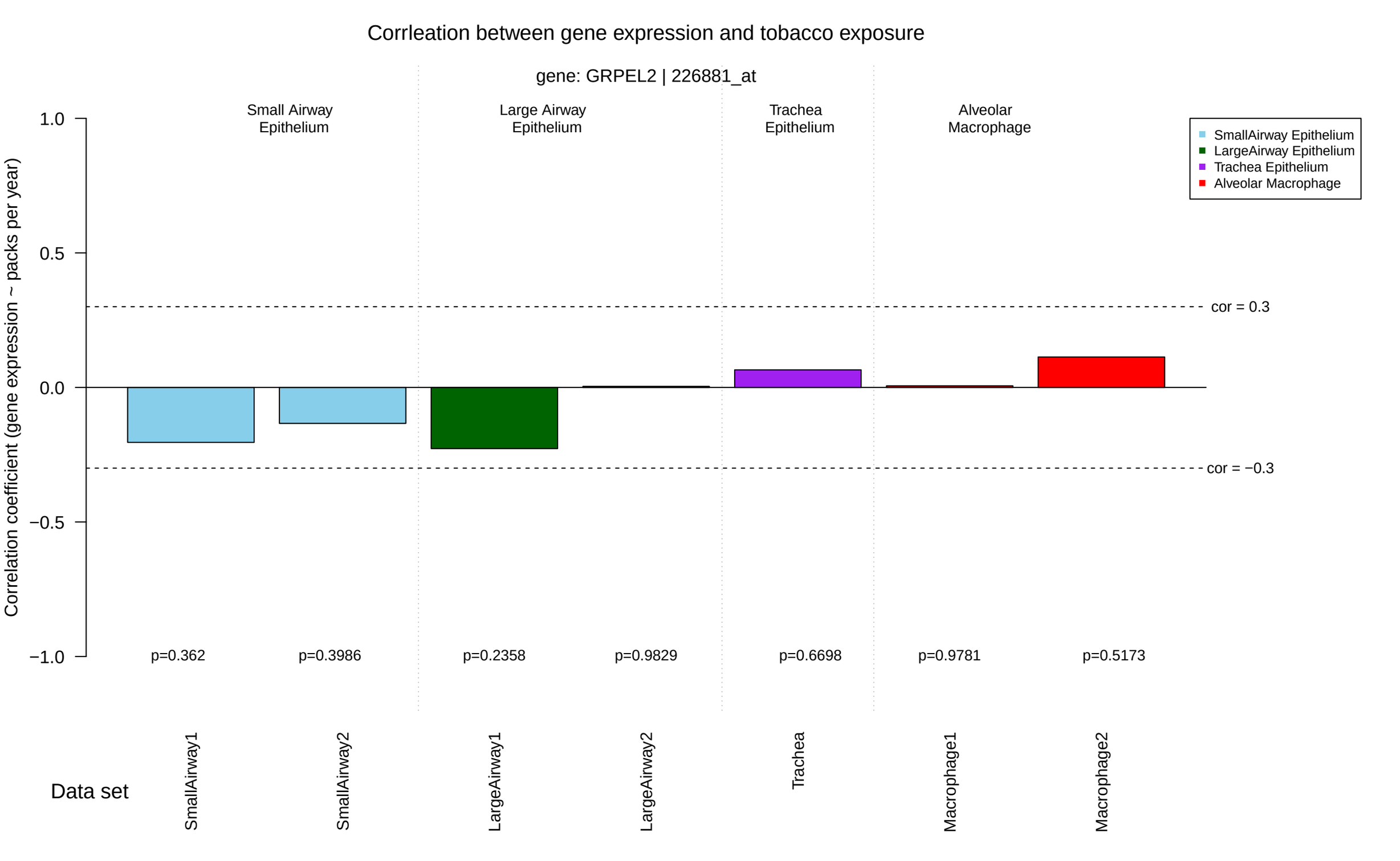

set: LargeAirway1 (p=0.0276)
GRPEL2 | 226881_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4484)
GRPEL2 | 226881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6779)
GRPEL2 | 226881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6925)
GRPEL2 | 226881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7001)
GRPEL2 | 226881_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRPEL2 | 226881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.362
p=0.3986
p=0.2358
p=0.9829
p=0.6698
p=0.9781
p=0.5173
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2